Arista Access Node MAR-ACC-45 Routing Protocol Flap SOP

Node: MAR-ACC-45 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_2499

Description

This Standard Operating Procedure (SOP) addresses a critical failure scenario on the Arista Access Node MAR-ACC-45 at the Marathahalli Core site, where a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) triggers a Warning severity Alarm (ALARM_2499). This SOP is designed to expedite the troubleshooting and resolution process by leveraging the diagnostic capabilities of Arista EOS. The underlying protocol mechanics driving this failure domain can be attributed to the complex interactions between the BGP, OSPF, and ISIS routing protocols, which, when misconfigured or experiencing issues, can lead to a cascading effect of routing protocol flaps. The Access role on the Arista node is particularly susceptible to these issues due to its position at the network's edge, where it is responsible for interconnecting with multiple upstream and downstream networks. The Arista EOS platform, being a software-defined networking solution, provides a rich set of diagnostic and troubleshooting tools that can be leveraged to identify and resolve these issues. By following this SOP, network administrators can efficiently isolate and resolve the root cause of the Routing Protocol Flap, minimizing downtime and ensuring the overall reliability of the network.

Pre-requisite Checks

- Verify that the Marathahalli Core site is accessible and that remote hands are dispatched to the location.

- Check the status of the redundant links to ensure that they are functioning correctly.

- Review the network topology and routing configurations to identify any potential issues or misconfigurations.

Potential Root Causes

Cause	Probability
BGP route churn due to misconfigured routing policies	High
OSPF adjacency loss due to incorrect interface configurations	Medium
ISIS LSP flooding due to hardware or software issues	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP table version 2 Local router ID: 10.1.1.1 Status codes: s-suppressed, d-damped, h-hold, *-valid, >-best, i-internal, r-RIB-failure, S-stale Origin codes: i - IGP, e - EGP, ? - incomplete Network Next Hop Metric LocPrf Weight Path *> 10.1.1.0/24 10.1.2.2 0 100 0 i, Description: This command displays a summary of the BGP table, including the local router ID,
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status codes, origin codes, and the best path to each network.

Command: show ospf database, Expected Output: OSPF database (Area 0)
Type ID Adv Rtr Seq Age LinkID Link Data Router 10.1.1.1 10.1.2.2
0x80000003 1200 10.1.2.2 10.1.3.3 Network 10.1.1.0 10.1.2.2 0x80000002
600 10.1.2.2 10.1.3.3, Description: This command displays the OSPF
database, including the type of LSA, ID, advertising router, sequence
number, age, link ID, and link data.

Command: show isis database, Expected Output: ISIS database (Level 2)
IS-IS LSP: 10.1.1.1/48 IS-IS LSP: 10.1.2.2/48, Description: This
command displays the ISIS database, including the level, LSP, and the
routers that are advertising the LSP.
```

Resolution Steps

Step 1: Step: Verify BGP routing policies, Command: show bgp policy, Description: This command displays the BGP routing policies, including the policy name, prefix, and the next hop. Verify that the policies are correctly configured and that there are no misconfigurations that could be causing the BGP route churn.

Step 2: Step: Verify OSPF interface configurations, Command: show ospf interface, Description: This command displays the OSPF interface configurations, including the interface name, IP address, and the OSPF process ID. Verify that the interface configurations are correct and that there are no issues with the OSPF process.

Step 3: Step: Verify ISIS LSP flooding, Command: show isis lsp, Description: This command displays the ISIS LSP flooding, including the LSP ID, router ID, and the level. Verify that the LSP flooding is not excessive and that there are no issues with the ISIS process.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP table version 2 Local
router ID: 10.1.1.1 Status codes: s-suppressed, d-damped, h-hold,
*-valid, >-best, i-internal, r-RIB-failure, S-stale Origin codes: i -
IGP, e - EGP, ? - incomplete Network Next Hop Metric LocPrf Weight
Path *> 10.1.1.0/24 10.1.2.2 0 100 0 i, Description: This command
displays a summary of the BGP table, including the local router ID,
status codes, origin codes, and the best path to each network. Verify
that the BGP routing policy has been correctly configured and that
there are no issues with the BGP process.

Command: show ospf database, Expected Output: OSPF database (Area 0)
Type ID Adv Rtr Seq Age LinkID Link Data Router 10.1.1.1 10.1.2.2
0x80000003 1200 10.1.2.2 10.1.3.3 Network 10.1.1.0 10.1.2.2 0x80000002
600 10.1.2.2 10.1.3.3, Description: This command displays the OSPF
database, including the type of LSA, ID, advertising router, sequence
number, age, link ID, and link data. Verify that the OSPF interface
configurations have been correctly configured and that there are no
issues with the OSPF process.

Command: show isis database, Expected Output: ISIS database (Level 2)
IS-IS LSP: 10.1.1.1/48 IS-IS LSP: 10.1.2.2/48, Description: This
command displays the ISIS database, including the level, LSP, and the
routers that are advertising the LSP. Verify that the ISIS LSP
flooding has been resolved and that there are no issues with the ISIS
process.
```

Roll-back Procedures

- Step: Restore the original BGP routing policy, Command: configure replace /path/to/backup/config, Description: This command restores the original BGP routing policy from a backup configuration. This should be done if the fix fails and the BGP routing policy needs to be reverted to its original state.

- Step: Restore the original OSPF interface configurations, Command: configure replace /path/to/backup/config, Description: This command restores the original OSPF interface configurations from a backup configuration. This should be done if the fix fails and the OSPF interface configurations need to be reverted to their original state.

- Step: Restore the original ISIS LSP flooding configurations, Command: configure replace /path/to/backup/config, Description: This command restores the original ISIS LSP flooding configurations from a backup configuration. This should be done if the fix fails and the ISIS LSP flooding configurations need to be reverted to their original state.